Does a Propaganda Detector Read Manipulation, or Just Vocabulary? A Construct-Validity Programme in Defensive Political Anthropology

An overview of the Politomorphism defensive-detection research line

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Code & data: github.com/politomorphism/politomorphism-defense

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Abstract

A machine that flags political text as propaganda may be reading the rhetorical structure that makes a passage manipulative, or merely the charged vocabulary that happens to accompany it in its training corpus. The two cannot be told apart on the corpus itself, where structure and vocabulary are confounded. This article gives an accessible overview of a small, self-funded research programme that tries to separate them, carried out as the defensive audit layer of the Politomorphism framework. It summarizes three working papers — a leakage-corrected lexical baseline, a fair comparison showing that context helps at equal review cost, and a contrastive-pairs diagnostic that isolates structure from vocabulary — and reports their combined result: on vocabulary-matched sentence pairs a lexical model performs near chance while a contextual model does not, a gap that is statistically significant, stable across training seeds, confirmed by an independent blind rater, and uncorrelated with residual vocabulary. It then describes work on turning detection into explanation (naming the device, not just scoring it), including a frozen configuration that met pre-registered acceptance criteria on a fresh author-built held-out set — a bounded result that is explicitly not independent validation. The whole programme is released openly for inspection and criticism.

1. A detector that reads words is not a detector of manipulation

Political manipulation is a matter of structure, not vocabulary. A text manipulates when it presupposes a contested claim as settled, forces a false either/or, deflects a criticism by pointing at the opponent, or asserts causation from mere sequence. The same charged words can appear in a plain, honest sentence; a manipulative structure can be built from mild ones. This matters for any tool meant to help audit political communication, because the easiest thing for a machine to learn is the vocabulary that correlates with propaganda in its training data, not the rhetorical mechanism. A detector can score well on a held-out slice of the same corpus while having learned only that lexical shortcut — and no evaluation performed on the corpus itself can reveal the difference, because there the words and the structures travel together.

This is the old question of construct validity: does an instrument measure what it claims to measure, or a confound? The programme described here was built to put that question to a propaganda detector directly, and to answer it honestly even when the answer was inconvenient.

2. A three-step programme, built in the open

The work proceeds in three working papers, each with code and data released publicly. The first establishes a transparent lexical baseline for detecting propagandistic paragraphs in the English

PropXplain corpus. Its main methodological point is defensive: the official corpus split leaks material from the same article across training and test, which inflates scores, so the baseline is rebuilt on an article-disjoint split. An earlier version of this work was published with that flaw and has since been corrected in the open — versioning and visible correction are treated here as marks of seriousness, not embarrassment.

The second paper compares that lexical baseline with a fine-tuned transformer on the same clean split, and makes a point about fair comparison. At their default thresholds the two models operate at different points, so a naive reading can mislead; judged at equal review cost and by threshold-independent measures, the contextual model is the better detector, and the advantage holds across five training seeds. But both papers remain, by their own admission, about predicting the corpus's labels — not about whether either model represents manipulation as a structure.

The third paper builds an instrument designed to answer exactly that. Its method is contrastive pairs.

3. The contrastive-pairs test, and what it showed

Each item is a pair of short political sentences on the same topic, written to share as many words as possible, where one member performs a single rhetorical device — presupposition, manufactured equivalence, false dilemma, whataboutism, a loaded rhetorical question, or implied causation — and the other states the same content plainly. Because the vocabulary is held nearly fixed, a model that scores the manipulative member higher must be responding to structure. On sixty such pairs, ten per device, the result points one way, and it is worth stating as five independent checks that agree:

First, on matched vocabulary the lexical baseline sits near the chance value of one-half (pair accuracy 0.63), while the contextual model does not (0.92). Second, because both models are scored on the identical pairs, the difference can be tested as a paired comparison, and it is significant (exact McNemar $p = 0.0002$). Third, the contextual model's margin is stable across five training seeds (mean 0.88, confidence interval excluding the baseline). Fourth, an independent rater that took no part in building the pairs — a separate language model, judging each sentence blind — recovers the manipulative/plain distinction with substantial agreement (Cohen's kappa 0.62). Fifth, the contextual model's advantage is uncorrelated with how much residual vocabulary distinguishes the two members, whereas the lexical model's success is strongly correlated with it — direct evidence that the contextual model wins on structure, not on leftover words. The gap is largest exactly where the lexical model fails structurally: false dilemma and loaded rhetorical questions.

The claim is deliberately narrow. This shows that a contextual model can be sensitive to rhetorical structure on a controlled instrument; it does not show that the model, or the corpus it learned from, captures manipulation as it operates in the world. Construct validity is earned incrementally, and this is one increment, not a verdict.

4. From detection to explanation, and a held-out check

A score is not yet useful to a human auditor. A further step, now underway, is explanation: naming which device a passage uses, so a reader can see the mechanism rather than trust a number. A zero-shot classifier names the device well when one is present but, in its first form, cried manipulation on plain text in about two-thirds of cases, keying on charged words. Adding the right to abstain and an instruction to respect negation and quotation cut that over-flagging by more than half without collapsing detection — though, tellingly, it did not raise the count of exactly-correct answers: it redistributed errors into abstentions, which for a defensive tool is a better failure, not a demonstrated gain. A targeted check for one failure mode — comparisons that are explicitly negated — removed those false alarms almost completely.

Because those numbers came from the same pairs that shaped the prompts and rules, and repeated tuning against the same examples can make results look too optimistic even without memorization, the configuration was frozen and run, unchanged, on a fresh author-built held-out set of sixty new pairs on different topics, with the acceptance thresholds fixed in advance. It met them: 16 false alarms out of 60 controls (threshold ≤ 22) and 54 correct detections out of 60 positives (threshold ≥ 48); the negation check then cut the negated-comparison false alarms from seven to zero, with no correct detection lost and no change outside the targeted class, so total false alarms fell from 16 to 9 while recall held at 90%. This is a bounded result: the frozen configuration met pre-set criteria on examples separate from development. It is not independent validation and does not demonstrate broad generalization, because the data and labels are the author's; independent human annotation remains future work. The honest status of the explanation layer is therefore a promising result that passed a pre-registered held-out check, not a settled one.

5. Ethics and scope

An instrument that scores rhetorical structure must never be used to decide which political positions are legitimate. A false dilemma or a loaded question can serve any cause; detecting the device says nothing about the truth or worth of the position it advances. The defensive purpose of this programme — to measure and explain manipulation, and to help readers see it, rather than to rank speakers or suppress speech — depends on keeping that line bright. The tool is meant to augment human judgment, not replace it: it flags and explains; a person decides. A detector that over-flags, or that is turned to adjudicate legitimacy, becomes the very thing it was built to resist, which is why the work dwells as much on false alarms and honest scope as on accuracy.

6. Open materials and an invitation

All of it — the three pair sets, the scoring harness, the per-sentence scores for the baseline and for every transformer seed, the independent rater's blind judgments, the explanation-layer scripts and their predictions, and the full protocols — is released in the public repository so that others can inspect, criticize, and extend it. The programme is the work of a single independent researcher, and it is the stronger for being contestable in detail. Readers are invited to break the pairs, to annotate them independently, to run the detectors on their own political text, and to say where the instrument fails. That is how construct validity is earned: rater by rater, example by example, in the open.

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